

# Mathematical Foundations of Political Methodology

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# Central Limit Theorem

- Moment Generating Functions
- Limit Theorems
- Properties of Estimators

# Approximating functions and second order conditions

## Theorem

**Taylor's Theorem** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x)$  is infinitely differentiable function. Then, the Taylor expansion of  $f(x)$  around  $a$  is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$$

## Example Function

Suppose  $a = 0$  and  $f(x) = e^x$ . Then,

$$\begin{aligned}f'(x) &= e^x \\f''(x) &= e^x \\&\vdots \\f^n(x) &= e^x\end{aligned}$$

This implies

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots + \frac{x^n}{n!} + \dots$$

# Moment Generating Functions

## Definition

Suppose  $X$  is a random variable with pdf  $f$ . Define,

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x) dx$$

We will call  $X^n$  the  $n^{\text{th}}$  moment of  $X$

- By this definition  $\text{var}(X) = \text{Second Moment} - \text{First Moment}^2$
- We are assuming that the integral converges

# Moment Generating Functions

## Proposition

Suppose  $X$  is a random variable with pdf  $f(x)$ . Call  $M(t) = E[e^{tX}]$ ,

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \end{aligned}$$

We will call  $M(t)$  the moment generating function, because:

$$\left. \frac{\partial^n M(t)}{\partial^n t} \right|_0 = E[X^n]$$

(Assuming that we can interchange derivative and integral)

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Recall the Taylor Expansion of  $e^{tX}$  at 0,



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$$E[e^{tX}] = 1 + tE[X] + \frac{t^2}{2!}E[X^2] + \frac{t^3}{3!}E[X^3] + \dots$$



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Differentiate once:

$$\begin{aligned}\frac{\partial M(t)}{\partial t} &= 0 + E[X] + \frac{2t}{2!}E[X^2] + \dots \\ M'(0) &= 0 + E[X] + 0 + 0 \dots\end{aligned}$$



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$$\frac{\partial^n M(t)}{\partial^n t} = 0 + 0 + 0 + \dots + \frac{n \times n - 1 \times \dots 2 \times t^0 E[X^n]}{n!} + \frac{n! t E[X^{n+1}]}{(n+1)!} + \dots$$



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- If two random variables,  $X$  and  $Y$  have the same moment generating functions, then  $F_X(x) = F_Y(y)$  for **almost all**  $x$ .

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## Proposition

*Suppose  $X_i$  are a sequence of independent random variables. Define*

$$Y = \sum_{i=1}^N X_i$$

*Then*

$$M_Y(t) = \prod_{i=1}^N M_{X_i}(t)$$

Proof.

$$M_Y(t) = E[e^{tY}]$$



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# Sequences and Convergence

- We use the concept of convergence to describe limits, derivatives and integrals in the first half of the course.
- Today we will discuss different kinds of convergence for random variables.
  - Convergence in distribution.
  - Convergence in probability.
  - Convergence almost surely.
  - Convergence surely or pointwise.

# Sequences and Convergence

Sequence (refresher):

$$\{a_i\}_{i=1}^{\infty} = \{a_1, a_2, a_3, \dots, a_n, \dots, \}$$

## Definition

*We say that the sequence  $\{a_i\}_{i=1}^{\infty}$  converges to real number  $A$  if for each  $\epsilon > 0$  there is a positive integer  $N$  such that for  $n \geq N$ ,  $|a_n - A| < \epsilon$*

# Convergence Pointwise

Sequence of functions:

$$\{f_i\}_{i=1}^{\infty} = \{f_1, f_2, f_3, \dots, f_n, \dots, \}$$

## Definition

Suppose  $f_i : X \rightarrow \mathfrak{R}$  for all  $i$ . Then  $\{f_i\}_{i=1}^{\infty}$  converges *pointwise* to  $f$  if, for all  $x \in X$  and  $\epsilon > 0$ , there is an  $N$  such that for all  $n \geq N$ ,

$$|f_n(x) - f(x)| < \epsilon$$

We would then say

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \forall x.$$

# Convergence of sequence of Random Variables

Define  $\hat{\theta}_n$  to be estimator for  $\theta$  based on  $n$  observations.

Question: What can we say about  $\{\hat{\theta}_i\}_{i=1}^n$  as  $n \rightarrow \infty$ ?

- Does  $\hat{\theta}_n(\omega)$  get arbitrarily close to  $\theta(\omega)$  for all  $\omega$ ?
- Does  $\hat{\theta}_n(\omega)$  get arbitrarily close to  $\theta(\omega)$  for all  $\omega$  that have positive probability?
- Does the probability that  $\hat{\theta}_n$  differs from  $\theta$  go to 0?
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# Convergence in Probability

## Definition

We will say the sequence  $\hat{\theta}_n$  converges **in probability** to  $\theta$  if,

$$\lim_{n \rightarrow \infty} \text{Prob}(|\hat{\theta}_n - \theta| > \epsilon) = 0$$

For any  $\epsilon > 0$

- The probability of an 'unusual' outcome becomes smaller and smaller as the sequence progresses.
- $\epsilon$  is a tolerance parameter: how much error around  $\theta$ ?
- In the limit, convergence in probability implies sampling distribution collapses on a spike at  $\theta$ .

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# Example of Convergence in Probability

- Toss a fair coin  $n$  times. Let  $R_n$  be the ‘longest run of heads’.
- So if  $n = 15$  and our experiment gets, HHTTHHHTHTHTHHH,  $R_{15} = 3$ .
- As  $n \rightarrow \infty$ ,  $\frac{R_n}{\log_2 n} \rightarrow 1$
- Plugging in 20 for  $R_n$ , we find that  $n = 2^{20} = 1048576$ .

## Example of Convergence in Probability (Consistency of $S^2$ )

- An estimator is called consistent if it converges in probability to the quantity being estimated.
- $S^2$  is the sample analogue of  $\sigma^2$ .

$$S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$$

- Suppose we have iid random variables  $E(X_i) = \mu$  and  $\text{Var}(X_i) = \sigma^2$
- Chebychev tells us that

$$P(|S^2 - \sigma^2| \geq \epsilon) \leq \frac{E(S_n^2 - \sigma^2)^2}{\epsilon^2} = \frac{\text{Var}S_n^2}{\epsilon^2}$$

- So  $S^2$  converges in probability to  $\sigma^2$  so long as  $\text{Var}S_n^2 \rightarrow 0$  as  $n \rightarrow \infty$

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- Suppose we have iid random variables  $E(X_i) = \mu$  and  $\text{Var}(X_i) = \sigma^2$
- Chebychev tells us that

$$P(|S^2 - \sigma^2| \geq \epsilon) \leq \frac{E(S_n^2 - \sigma^2)^2}{\epsilon^2} = \frac{\text{Var}S_n^2}{\epsilon^2}$$

- So  $S^2$  converges in probability to  $\sigma^2$  so long as  $\text{Var}S_n^2 \rightarrow 0$  as  $n \rightarrow \infty$

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Except for a subset  $\mathcal{N} \subset S$  such that  $P(\mathcal{N}) = 0$ .

## Example Almost Sure

- Consider an animal of some short-lived species. We record the amount of food that this animal consumes per day. This sequence of numbers will be unpredictable, but we may be quite certain that one day the number will become zero, and will stay zero forever after.

## Example (Cassella and Berger)

Suppose  $S \sim \text{Uniform}(0,1)$ .

Suppose  $X_n$  is define as follows:

$$X_1(s) = s + I(s \in [0, 1]) \quad , \quad X_2(s) = s + I(s \in [0, 1/2])$$

$$X_3(s) = s + I(s \in [1/2, 1]) \quad , \quad X_4(s) = s + I(s \in [0, 1/3])$$

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**No!**: the sequence doesn't converge for each  $s$

For each value of  $s$  the sequence varies between  $s$  and  $s + 1$  infinitely often

# Convergence in Distribution

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$$\lim_{n \rightarrow \infty} F_n(x) - F(x) = 0$$

*For all  $x \in \mathbb{R}$  where  $F(x)$  is continuous.*

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- Weakest form of convergence almost sure  $\rightarrow$  probability  $\rightarrow$  distribution
- Says that cdfs are nearly the same, says nothing about convergence of underlying RV
- Useful for justifying use of some sampling distributions

# Example of Convergence in Distribution

- Suppose a new dice factory has just been built. The first few dice come out quite biased, due to imperfections in the production process. The outcome from tossing any of them will follow a distribution markedly different from the desired uniform distribution.
- As the factory is improved, the dice become less and less loaded, and the outcomes from tossing a newly produced die will follow the uniform distribution more and more closely.

# Example of Convergence in Distribution

- Suppose that  $X$  is a continuous random variable with range  $\mathbb{X}_n \equiv (0, n]$  for  $n > 0$  and cdf:

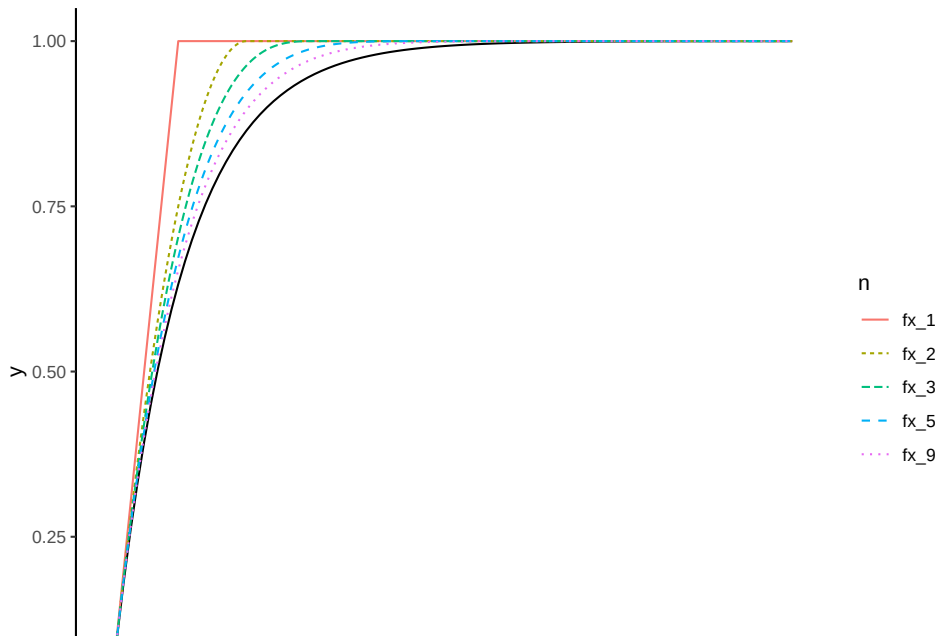
$$F_{X_n}(x) = 1 - \left(1 - \frac{x}{n}\right)^n, \quad 0 < x \leq n$$

- As  $n \rightarrow \infty$ ,

$$F_{X_n}(x) \rightarrow 1 - e^{-x}$$

- So  $X_n \xrightarrow{d} X$ ,  $X \sim \text{Exponential}(1)$

# Example of Convergence in Distribution



# Convergence in Distribution $\not\Rightarrow$ Convergence in Probability

Define  $X \sim N(0, 1)$  and each  $X_n = -X$ . Then:  
 $X_n \sim N(0, 1)$  for all  $n$  so  $X_n$  trivially converges to  $X$ . But,

$$\begin{aligned} P(|X_n - X| > \epsilon) &= P(|X + X| > \epsilon) \\ &= P(|2X| > \epsilon) \\ &= P(|X| > \epsilon/2) \not\rightarrow 0 \end{aligned}$$

# Central Limit Theorem

## Proposition

Let  $X_1, X_2, \dots$  be a sequence of independent random variables with mean  $\mu$  and variance  $\sigma^2$ . Let  $X_i$  have a cdf  $P(X_i \leq x) = F(x)$  and moment generating function  $M(t) = E[e^{tX_i}]$ . Let  $S_n = \sum_{i=1}^n X_i$ . Then

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n - \mu n}{\sigma \sqrt{n}} \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{z^2}{2}\right) dz$$

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Proof plan:

- 1) Rely on Fact that convergence of MGFs  $\rightsquigarrow$  convergence in CDFs
- 2) Show that MGFs, in limit, converge on normal MGF

## Proposition

*Let  $F_n$  be a sequence of cumulative distribution functions with corresponding moment generating functions  $M_n$ .  $F$  be a cdf with the moment generating functions  $M$ . If  $\lim_{n \rightarrow \infty} M_n(t) \rightarrow M(t)$  for all  $t$  in some interval, then  $F_n(x) \rightsquigarrow F(x)$  for all  $x$  (when  $F$  is continuous).*

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## Proposition

Suppose  $M(t)$  is a moment generating function of some random variable  $X$ . Then  $M(0) = 1$ .

# Proof of Central Limit Theorem

Proof. Suppose  $X_1, \dots, X_n$  are i.i.d. variables with  $E[X] = 0$ , variance  $\sigma_x^2$ , Moment Generating Function (MGF)  $M_x(s)$ .

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Using Taylor's Theorem we can write

$$M_x(s) = M_x(0) + sM'_x(0) + \frac{1}{2}s^2M''_x(0) + e_s$$

$$e_s/s^2 \rightarrow 0 \text{ as } s \rightarrow 0.$$

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Filling in the derivatives of  $M_x$  we have

$$M_x(s) = 1 + 0 + \frac{1}{2}s^2\sigma_x^2 + \underbrace{e_s}_{\text{Goes to zero}}$$

Using the fact that  $M_{Z_n}(s) = (M_x(\frac{s}{\sigma_x\sqrt{n}}))^n$

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# Estimators

- Statistics are functions of random variables.
- An *estimator* is statistic that is used to infer the value of an unknown parameter.
- An estimate is a particular realization of the estimator.
- Estimators are written with a  $\hat{\theta}$ .
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# Properties of Estimators

- Estimators have distributions, expectations and variances.
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# Bias

The bias of an estimator is defined to be

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- If  $E(\hat{\theta}) > \theta$ , the estimator is biased upward.
- If  $E(\hat{\theta}) < \theta$ , the estimator is biased downward.
- If  $E(\hat{\theta}) = \theta$ , the estimator is unbiased.

# Example of Unbiased Estimator

- Sample mean  $E(\bar{X}) - \mu$

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$$\begin{aligned} E[\hat{\sigma}^2] &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2\right] \\ &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i^2) - \frac{1}{n} 2\bar{X}_n \sum_{i=1}^n X_i + \frac{1}{n} \sum_{i=1}^n (\bar{X}_n^2)\right] \\ &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i^2) - \frac{1}{n} 2\bar{X}_n n\bar{X}_n + \frac{n}{n} \bar{X}_n^2\right] \\ &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i^2) - \bar{X}_n^2\right] \\ &= \frac{1}{n} \sum_{i=1}^n E[X_i^2] - E[\bar{X}_n^2] \\ &= \frac{1}{n} \sum_{i=1}^n [\sigma^2 + \mu^2] - \left[\frac{\sigma^2}{n} + \mu^2\right] \end{aligned}$$

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# Efficiency

Given two unbiased estimators, we can compare their sampling error.

$$\sigma_{\hat{\theta}}^2 = E[(\hat{\theta} - E(\hat{\theta}))^2]$$

- Estimator  $\hat{\theta}$  is more efficient than  $\tilde{\theta}$  if  $\sigma_{\hat{\theta}} < \sigma_{\tilde{\theta}}$
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# Example of Efficiency

- 'First Impression' Estimator:  $\mathcal{W}(X_1, X_2, X_3, \dots) = X_1$

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# Mean squared Error

- If we have a biased estimator, we can compare sampling error by evaluating *mean squared error*

$$\text{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

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# Design of Studies with a known $\mu$ and $\sigma$

- How many draws should we get to make sure that our sample mean is close to the truth?
- Suppose that we assume that our data is normal, with a known mean  $\mu$  and  $\sigma^2$
- What is

$$P((\bar{X}_n - \mu)/(\sigma/\sqrt{n}) < z | \mu, \sigma^2, n)$$

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# Design of Studies

- All we need to do is find the smallest  $n$  that produces:

$$P((\bar{X}_n - \mu)/(\sigma/\sqrt{n}) < z | \mu, \sigma^2, n) \leq \alpha$$

- And solve for  $n$ .

## Sample size when we know $\mu$ and $\sigma$

- Imagine that you are running a survey, and have to pick a sample size ahead of time such that

$$P(|\bar{X} - \mu| < \sigma/5) = .95$$

- What sample size would you choose?  $\Phi^{-1}(.95) = 1.644854$

$$\Phi^{-1}(.975) = 1.959$$

$$P\left(\frac{|\bar{X}_1 - \mu|}{\frac{\sigma}{\sqrt{n}}} < 1.96\right) = .95$$

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# Design of studies

- If you are asked, 'How large a sample size do I need'
- Ask:
- How accurately do you need the answer?
- What level of confidence do you intend to use?
- What is your current estimate?

# Finite Sample Properties of Estimators

- If  $X_i \sim N(\mu, \sigma^2)$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

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